

Dexter Fernandes

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https://github.com/Dexter-Fernandes

Computer Vision Engineer

Computer Vision Engineer with an MSc in Robotics, experienced in developing production-ready systems using Python, OpenCV, PyTorch and TensorFlow. At LivNSense, I improved CNN model precision by 33% and increased throughput to 15 FPS by rearchitecting the inference pipeline for real-time inference. I have designed and run end-to-end training and inference pipelines, deploying optimised models to edge devices such as Jetson and Intel VPUs using TensorRT and OpenVINO. I have also built and maintained robust application backends with Django and PostgreSQL, containerised with Docker within CI/CD workflows. I have used MLflow to build robust MLOps pipelines to drive systematic, data-backed performance improvements.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed real-time perception modules for oil-rig safety systems, deploying CNN-based vision pipelines with YOLOv5 on edge devices to improve detection accuracy by 15%.
- Designed automated data collection, augmentation, and labeling tools to scale large image datasets, increasing dataset diversity by 40% and cutting manual labeling time by 60%.
- Built automated **training and validation pipelines** integrated with **MLflow** for experiment tracking and **Optuna** for Bayesian hyperparameter tuning, cutting model iteration time by **40%**.
- Refactored legacy monolith into Kafka-driven microservices, containerised with Docker for fault isolation and 35% higher MTBF.
- Integrated AI modules with production systems (Docker, FastAPI, Kafka, Git CI/CD) in collaboration with software/hardware/QA teams to deliver fault-tolerant edge deployments.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60%.
- Implemented rigorous testing and validation frameworks simulating challenging lighting, occlusion, and motion conditions, improving CNN stability and reducing variance by 25%.
- Collaborated with software, hardware and QA teams to integrate AI modules into **Docker/FastAPI** production systems using **Git CI/CD**, ensuring reproducible, fault-tolerant deployments.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and RGB data for teleoperation of an Unmanned Ground Vehicle (UGV).
- Processed and analysed large-scale point cloud datasets using Open3D and RTAB-Map to perform 3D feature extraction, registration, and map optimization.
- Tuned SLAM parameters to improve localisation accuracy across varied terrains, achieving stable performance in both feature-dense indoor and feature-sparse outdoor environments.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS 2 before successful real-world deployment.
- Integrated multiple sensing modalities to achieve robust spatial and radiological understanding, enabling accurate obstacle detection in static environments.
- Implemented waypoint navigation in simulation using ROS2 Nav2 (standard planners/controllers) within Gazebo, validating end-to-end navigation workflows.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 2D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Deep Learning and Optimisation:** Convolutional Neural Networks, DeepSORT, OpenVINO, TensorRT, ONNX, Optuna, MultiRocket, Bayesian Optimisation, PyTorch, TensorFlow
- **Perception & SLAM:** Vision Transformer (ViT), OpenCV, LiDAR SLAM, Open3D, Point Cloud Library, 3D Feature Extraction, Scan Matching, Denoising, NumPy, Scikit-Image, Data Augmentation, YOLO, U-Net, SAM
- **Cloud, Edge & MLOps:** AWS, Microsoft Azure, MLFlow, Intel VPUs, NVIDIA Jetson, FastAPI, Apache Kafka, DVC (Data Version Control), Docker, Linux, Git
- **Programming & Development:** Python, C++, Bash, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy), Gazebo (Harmonic), Nav2, Pandas, PostgreSQL, Django, Qt, CUDA